



12+ page PDF handbook

HIVE MONITORING COURSE

A technical handbook for reading BeeYield sensor signals and turning alerts into field action.

Who this is for

Apiary managers, field technicians, and operations staff using BeeYield sensors or similar monitoring systems.

Guide summary

Monitoring only creates value when teams know what the signal means and what action follows. This handbook explains the operational logic behind temperature, humidity, weight, location, and acoustic signals, then links each metric to troubleshooting and deployment decisions.

BeeYield brand standard: every page of this PDF carries BeeYield branding, page numbering, and a repeatable working format so the guide can be used in the yard, classroom, or packing room.

How to use this guide

Read the main sections in order, then use the worksheets and tables as operational tools. This document is written for field use, so the recommended action is always more important than memorising definitions.

Learning outcomes

- Understand what the main hive signals represent biologically and operationally.
- Install and maintain sensors so the data stream stays useful.
- Interpret abnormal readings without jumping to the wrong conclusion.
- Use alerts to prioritise field visits, not to replace inspection judgement.

Core Monitoring Signals

Temperature	Brood nest regulation and colony stability	Is brood healthy and is the colony maintaining the core?
Humidity	Ventilation, nectar drying, and internal moisture	Is the hive too wet, too crowded, or poorly ventilated?
Weight	Nectar flow, stores, and sudden disturbances	Is the colony bringing in food or losing stores unusually fast?
Acoustics	Colony mood and queen-related or swarming signals	Is the colony operating normally or shifting into stress?
GPS / movement	Location integrity and theft risk	Has the hive moved when it should not have moved?
Battery / connectivity	Sensor health	Can the data still be trusted and retrieved on time?

1. Monitoring is useful when it changes decisions

A dashboard is not proof of better management by itself. Monitoring becomes valuable when it helps a team decide which hive needs attention first, which yard is ready for harvest, which deployment is losing strength, or which alert can safely wait. Without a response workflow, the system turns into theatre.

BeeYield's own product language makes this point clearly: sensors are there to create visibility and accountability. That means every monitored metric should be tied to a management question and to a responsible person who owns the next action.

In a healthy operation, monitoring does not replace inspections. It tells the team where to inspect sooner, what to expect, and which risks are already visible before lids come off.

Field actions

- Connect every signal to a management decision.
- Define who responds to each alert category.
- Use data to prioritise work, not to create noise.
- Keep field verification in the loop.

2. Temperature and humidity reveal the brood environment

Brood rearing depends on a narrow thermal and moisture environment. Stable internal temperature usually reflects a colony with enough bees to regulate the brood nest. When the signal becomes unstable, the cause may be colony weakening, brood movement, extreme weather, or sensor placement error. The data should trigger a hypothesis, not a conclusion.

Humidity adds context. High internal moisture may point to nectar processing, poor ventilation, crowding, or wet seasonal conditions. Low humidity can sometimes be normal during strong ventilation or dry periods. The field team should therefore interpret temperature and humidity together and against recent weather.

A single spike matters less than a pattern. Healthy monitoring reads trends over time and compares one hive with its neighbours before escalating.

Field actions

- Read temperature and humidity together.
- Investigate patterns, not isolated moments, first.
- Check sensor placement before assuming colony failure.
- Compare a suspicious hive with nearby monitored colonies.

3. Weight data turns nectar flow into something operational

Hive weight has long been one of the clearest indicators of nectar flow and store movement. Daily gains can show that a bloom is working well, while steep unexpected drops may indicate dearth, robbing, harvest removal, or a disturbance that needs investigation. Weight also helps operations plan supers and transport more accurately.

The value of weight data rises when it is read in context. Rain, equipment movement, and handling can create sudden steps in the curve that have nothing to do with forage. Teams should log those events so analysts do not mistake management activity for bee activity.

When combined with bloom records, weight curves become especially useful for honey operations and precision pollination jobs because they show whether bees are converting field opportunity into stored resources.

Field actions

- Use weight trends to time supering and harvest planning.
- Log handling events that affect the scale trace.
- Investigate unexplained drops quickly.
- Pair weight with bloom notes to interpret forage conditions honestly.

4. Acoustic monitoring is powerful when treated carefully

Sound can reveal shifts in colony state that are hard to see from outside the hive. Acoustic systems may help detect queenlessness, swarming mood, or agitation, but they should be treated as a signal layer that prompts inspection rather than a final diagnosis. Colonies are biologically noisy, and many sounds overlap.

The operational advantage is speed. If a monitored yard produces unusual acoustic behaviour at the same time that temperature stability drops or activity declines, the combined evidence helps managers prioritise a field visit. Sound becomes more reliable when it is interpreted with other signals rather than in isolation.

Good acoustic monitoring also depends on consistent sensor position and a stable device mount. Poor placement creates noisy data that looks sophisticated but explains little.

Field actions

- Use sound as part of a signal stack, not as a standalone verdict.
- Keep microphone placement consistent across monitored hives.
- Escalate only when acoustic change is supported by other evidence.
- Review field notes after inspections to improve future alert interpretation.

5. Sensor health is part of hive health

A dead battery, loose connector, failed gateway, or shifted probe can create a false emergency or hide a real one. Monitoring teams therefore need a device maintenance routine that sits alongside bee work: battery checks, calibration checks, mounting checks, and communication checks.

The fastest way to lose trust in a monitoring system is to leave obvious device failures unresolved. Field staff stop believing alerts when they know the same hardware problem keeps repeating. Reliability depends as much on maintenance culture as on the sensors themselves.

Documenting device service also helps during troubleshooting. If a data gap starts after a battery replacement or a moved sensor, the investigation starts with the most likely cause instead of wasting an inspection.

Field actions

- Schedule sensor maintenance instead of waiting for failure.
- Log every reposition, battery change, and calibration check.
- Treat connectivity loss as an operations issue, not only an IT issue.
- Separate biological alerts from device alerts in reporting.

6. Build alert playbooks that crews can use under pressure

Monitoring systems are most useful when alert response is boringly clear. A heat collapse alert, a sudden weight drop, a theft movement alert, or a long connectivity gap should each have a short playbook: who checks it, what evidence they review first, when they call the field team, and what counts as closure.

This is where a course becomes an operations tool. Staff should not improvise every alert from scratch. They should open a playbook, verify the signal, inspect when needed, take action, and close the loop with notes. Over time, those notes become the local knowledge that makes alerts more accurate and response more economical.

Hive monitoring delivers its best value when data, field discipline, and communication all move together.

Field actions

- Write playbooks for the few alert types that matter most.
- Escalate quickly when multiple signals fail together.
- Close alerts with notes that teach the system something.
- Measure alert quality by response quality, not by alert volume.

Glossary

Telemetry	The remote collection and transmission of field measurements.
Baseline	The expected normal range for a hive or yard under known conditions.
Anomaly	A reading or pattern that departs meaningfully from the baseline.
Calibration	The process of confirming that a device reports accurately enough to trust.
False positive	An alert that looks serious in data but is not a real biological problem.
Signal drift	Gradual change caused by sensor ageing, positioning, or power issues rather than the bees.

Sensor Deployment Record

Log sensor ID, hive ID, placement location in the hive, install date, calibration note, battery state, and assigned technician.

Alert Response Log

Capture alert time, signal type, probable cause, field verification result, and the corrective action taken.

References

This guide was written from BeeYield operating practice and the following external references. Review them when adapting the guide to a new crop, region, or compliance requirement.

USDA ARS - Monitoring colony phenology using within-day variability in continuous weight and temperature of honey bee hives

<https://www.ars.usda.gov/research/publications/publication/?seqNo115=313374>

FAO / Bees for Development - Beekeeping and sustainability

<https://www.fao.org/family-farming/detail/en/c/1755592/>

BeeYield product and dashboard telemetry model

Internal BeeYield monitoring workflows and dashboard metrics